

Latent Variable Models

Lecture Notes

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1 Introduction to Latent Variable Models

1.1 Motivation

Latent variable models explain correlations in observed variables $\mathbf{x}$ by assuming dependence on unobserved (latent) variables $\mathbf{z}$. These latent variables can represent hierarchical structure:

- High-level: objects, illumination, pose
- Mid-level: object parts, surfaces
- Low-level: edges
- Observed level: retinal image (pixels)

Definition 1.1 (Latent Variable Model). A latent variable model consists of:

$$\mathbf{z} \sim P[\theta_z] \tag{1}$$

$$\mathbf{x} \mid \mathbf{z} \sim P[\theta_x] \tag{2}$$

The joint distribution is:

$$p(\mathbf{x}, \mathbf{z}; \theta_x, \theta_z) = p(\mathbf{x} \mid \mathbf{z}; \theta_x) p(\mathbf{z}; \theta_z) \tag{3}$$

The marginal distribution of observables is:

$$p(\mathbf{x}; \theta_x, \theta_z) = \int d\mathbf{z} p(\mathbf{x} \mid \mathbf{z}; \theta_x) p(\mathbf{z}; \theta_z) \tag{4}$$

1.2 Purposes of Latent Variable Models

- **Describe structured distributions:** Correlations in high-dimensional $\mathbf{x}$ may be captured by fewer parameters.
- **Capture underlying generative process:** $\mathbf{z}$ may describe causes of $\mathbf{x}$ and help separate signal from noise.
- **Combine exponential family distributions:** $P(\mathbf{z})$, $P(\mathbf{x}|\mathbf{z})$, and even $P(\mathbf{x}, \mathbf{z})$ may be in the exponential family, while $P(\mathbf{x})$ rarely is (Exception: Linear Gaussian models).

2 Latent Variables and Gaussian Distributions

2.1 Gaussian Correlation Decomposition

Example 2.1 (Gaussian Correlation via Latent Variables). Gaussian correlation can be composed from latent components and uncorrelated noise (cf. Definition 1.1):

$$\mathbf{x} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}\right) \Leftrightarrow \begin{cases} z \sim \mathcal{N}(0, 1) \\ \mathbf{x} \sim \mathcal{N}\left(\sqrt{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} z, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \end{cases} \tag{5}$$

3 Probabilistic Principal Components Analysis (PPCA)

3.1 Model Definition

Definition 3.1 (PPCA Model). If the uncorrelated noise is assumed to be isotropic, the model is called PPCA (a special case of Definition 1.1).

Data: $\mathcal{D} = \mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ with $\mathbf{x}_i \in \mathbb{R}^D$

Latents: $\mathcal{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N\}$ with $\mathbf{z}_i \in \mathbb{R}^K$

Linear generative model:

$$x_d = \sum_{k=1}^K \Lambda_{dk} z_k + \epsilon_d \quad (6)$$

where:

- z_k are independent $\mathcal{N}(0, 1)$ Gaussian factors
- ϵ_d are independent $\mathcal{N}(0, \psi)$ Gaussian noise
- $K < D$

3.2 Probabilistic Formulation

The model for observations $\mathbf{x}$ (Definition 3.1) is a correlated Gaussian:

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I) \quad (7)$$

$$p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\Lambda\mathbf{z}, \psi I) \quad (8)$$

Note 3.2 (Expectation Identities). For computing the marginal distribution, we use:

$$\mathbb{E}_{\mathbf{x}}[f(\mathbf{x})] = \mathbb{E}_{\mathbf{z}}[\mathbb{E}_{\mathbf{x}|\mathbf{z}}[f(\mathbf{x})]] \quad (9)$$

$$\text{Var}_{\mathbf{x}}[\mathbf{x}] = \mathbb{E}_{\mathbf{z}}[\text{Var}[\mathbf{x}|\mathbf{z}]] + \text{Var}_{\mathbf{z}}[\mathbb{E}[\mathbf{x}|\mathbf{z}]] \quad (10)$$

Proposition 3.3 (PPCA Marginal Distribution). *The marginal distribution of the PPCA model (Definition 3.1) is:*

$$p(\mathbf{x}) = \int p(\mathbf{z})p(\mathbf{x}|\mathbf{z})d\mathbf{z} = \mathcal{N}(\mathbf{0}, \Lambda\Lambda^T + \psi I) \quad (11)$$

where Λ is a $D \times K$ matrix. The computation uses the identities in Note 3.2.

3.3 Comparison: Two Gaussian Models

3.3.1 Descriptive Density Model

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{0}, \Sigma) \quad (12)$$

Properties:

- Correlations captured by off-diagonal elements of Σ
- Σ has $\frac{D(D+1)}{2}$ free parameters
- Only constrained to be positive definite
- Simple ML estimate

3.3.2 Interpretable Causal Model

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I) \quad (13)$$

$$p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\Lambda\mathbf{z}, \psi I) \quad (14)$$

$$\Rightarrow p(\mathbf{x}) = \mathcal{N}(\mathbf{0}, \Lambda\Lambda^T + \psi I) \quad (15)$$

Properties:

- Correlations captured by common influence of latent variable
- $\Lambda\Lambda^T + \psi I$ has $DK + 1$ free parameters
- For $K < D$, covariance structure is constrained (“blurry pancake”)
- ML estimation is more complex

3.4 PPCA Likelihood

Proposition 3.4 (PPCA Log-Likelihood). *The marginal distribution on $\mathbf{x}$ (Proposition 3.3) gives the PPCA likelihood:*

$$\log p(\mathcal{X}|\Lambda, \psi) = -\frac{N}{2} \log |2\pi(\Lambda\Lambda^T + \psi I)| - \frac{1}{2} \text{Tr} \left[(\Lambda\Lambda^T + \psi I)^{-1} \sum_n \mathbf{x}_n \mathbf{x}_n^T \right] \quad (16)$$

where we define $S = \frac{1}{N} \sum_n \mathbf{x}_n \mathbf{x}_n^T$.

Remark 3.5 (ML Optimization Strategies). To find the ML values of (Λ, ψ) , we could:

1. Optimize numerically (gradient ascent/Newton’s method)
2. Use the EM algorithm (introduced later)
3. Analyze the limit $\psi \rightarrow 0$ (leads to PCA, see Section 4)

Note 3.6 (Mean Parameter). We may add a constant mean $\boldsymbol{\mu}$ to the output to model data not distributed around $\mathbf{0}$. The ML estimate is $\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_n \mathbf{x}_n$, and we define:

$$S = \frac{1}{N} \sum_n (\mathbf{x}_n - \hat{\boldsymbol{\mu}})(\mathbf{x}_n - \hat{\boldsymbol{\mu}})^T \quad (17)$$

3.5 The $\psi \rightarrow 0$ Limit

Remark 3.7 (Limit to PCA). As $\psi \rightarrow 0$, the latent model (Definition 3.1) can only capture K dimensions of variance. In a Gaussian model, the ML parameters will find the K -dimensional space of most variance. This leads to Principal Components Analysis (PCA), discussed in Section 4.

4 Principal Components Analysis

4.1 Algorithm Description

Definition 4.1 (Principal Components Analysis). Assume data $\mathcal{D} = \{\mathbf{x}_i\}$ have zero mean (if not, subtract it).

1. Find direction of greatest variance $-\lambda_{(1)}$:

$$\lambda_{(1)} = \arg \max_{\|\mathbf{v}\|=1} \sum_n (\mathbf{x}_n^T \mathbf{v})^2 \quad (18)$$

2. Find direction orthogonal to $\lambda_{(1)}$ with greatest variance $-\lambda_{(2)}$
3. Find direction orthogonal to $\{\lambda_{(1)}, \lambda_{(2)}, \dots, \lambda_{(n-1)}\}$ with greatest variance $-\lambda_{(n)}$
4. Terminate when remaining variance drops below a threshold

4.2 Eigendecomposition of Covariance Matrix

Definition 4.2 (Eigenvector and Eigenvalue). $\mathbf{u}$ is an eigenvector, with scalar eigenvalue ω , of a matrix S if:

$$S\mathbf{u} = \omega\mathbf{u} \quad (19)$$

We normalize $\mathbf{u}$ to have unity norm: $\mathbf{u}^T \mathbf{u} = 1$.

Proposition 4.3 (Covariance Eigendecomposition). For a covariance matrix $S = \langle \mathbf{x}\mathbf{x}^T \rangle$ (which is $D \times D$, symmetric, positive semi-definite):

1. There are D eigenvector-eigenvalue pairs $(\mathbf{u}_{(i)}, \omega_{(i)})$ (as in Definition 4.2), except if eigenvalues are degenerate
2. The D eigenvectors are orthogonal (or orthogonalizable if $\omega_{(i)} = \omega_{(j)}$), forming an orthonormal basis:

$$\sum_i \mathbf{u}_{(i)} \mathbf{u}_{(i)}^T = I \quad (20)$$

3. Any vector $\mathbf{v}$ can be written as:

$$\mathbf{v} = \sum_i \mathbf{u}_{(i)} \mathbf{u}_{(i)}^T \mathbf{v} = \sum_i (\mathbf{u}_{(i)}^T \mathbf{v}) \mathbf{u}_{(i)} = \sum_i v_{(i)} \mathbf{u}_{(i)} \quad (21)$$

4. The original matrix S can be written:

$$S = \sum_i \omega_{(i)} \mathbf{u}_{(i)} \mathbf{u}_{(i)}^T = U W U^T \quad (22)$$

where $U = [\mathbf{u}_{(1)}, \mathbf{u}_{(2)}, \dots, \mathbf{u}_{(D)}]$ and $W = \text{diag}(\omega_{(1)}, \omega_{(2)}, \dots, \omega_{(D)})$

4.3 PCA and Eigenvectors

Proposition 4.4 (Variance in Eigenvector Direction). *The variance in direction $\mathbf{u}_{(i)}$ (Definition 4.2) is:*

$$\langle (\mathbf{x}^T \mathbf{u}_{(i)})^2 \rangle = \langle \mathbf{u}_{(i)}^T \mathbf{x} \mathbf{x}^T \mathbf{u}_{(i)} \rangle = \mathbf{u}_{(i)}^T \mathbf{S} \mathbf{u}_{(i)} = \mathbf{u}_{(i)}^T \boldsymbol{\omega}_{(i)} \mathbf{u}_{(i)} = \omega_{(i)} \quad (23)$$

Proposition 4.5 (Variance in Arbitrary Direction). *The variance in an arbitrary direction $\mathbf{v}$ is (using the expansion (21) from Proposition 4.3):*

$$\begin{aligned} \langle (\mathbf{x}^T \mathbf{v})^2 \rangle &= \left\langle \left(\mathbf{x}^T \sum_i v_{(i)} \mathbf{u}_{(i)} \right)^2 \right\rangle = \sum_{ij} v_{(i)} \mathbf{u}_{(i)}^T \mathbf{S} \mathbf{u}_{(j)} v_{(j)} \\ &= \sum_{ij} v_{(i)} \omega_{(j)} v_{(j)} \mathbf{u}_{(i)}^T \mathbf{u}_{(j)} = \sum_i v_{(i)}^2 \omega_{(i)} \end{aligned} \quad (24)$$

Theorem 4.6 (Maximum Variance Direction). *If $\mathbf{v}^T \mathbf{v} = 1$, then $\sum_i v_{(i)}^2 = 1$, and (by Proposition 4.5):*

$$\arg \max_{\|\mathbf{v}\|=1} \langle (\mathbf{x}^T \mathbf{v})^2 \rangle = \mathbf{u}_{(\max)} \quad (25)$$

where $\mathbf{u}_{(\max)}$ is the eigenvector with the largest eigenvalue.

Corollary 4.7 (Principal Components as Eigenvectors). *From Theorem 4.6, the principal components (Definition 4.1) are exactly the eigenvectors of the empirical covariance matrix, ordered by decreasing eigenvalue.*

Remark 4.8 (Eigenspectrum). The eigenspectrum shows how variance is distributed across dimensions. It can:

- Identify transitions separating signal from noise
- Determine number of PCs capturing a pre-determined fraction of variance

4.4 PCA Subspace

Definition 4.9 (PCA Projection). The K principal components (Corollary 4.7) define the K -dimensional subspace of greatest variance. Each data point $\mathbf{x}_n$ is associated with a projection $\hat{\mathbf{x}}_n$ into the principal subspace:

$$\hat{\mathbf{x}}_n = \sum_{k=1}^K x_{n(k)} \boldsymbol{\lambda}_{(k)} \quad (26)$$

This can be used for lossy compression, denoising, recognition, etc.

4.5 Examples of PCA

Example 4.10 (Eigenfaces). PCA (Definition 4.1) applied to facial images produces eigenvectors representing common facial features. The mean face and principal eigenfaces can be used for face recognition and reconstruction.

Example 4.11 (Genetic Variation within Europe). PCA applied to genetic data shows that the first two principal components closely correspond to geographic location (north-south and east-west axes in Europe). This demonstrates that genetic variation correlates with geographic distance.

Reference: Novembre et al. (2008) Nature 456:98-101

4.6 Equivalent Definitions of PCA

PCA can be equivalently defined as:

- Find K directions of greatest variance in data
- Find K -dimensional orthogonal projection that preserves greatest variance
- Find K -dimensional vectors $\mathbf{z}_i$ and matrix Λ so that $\hat{\mathbf{x}}_i = \Lambda \mathbf{z}_i$ is as close as possible (in squared distance) to $\mathbf{x}_i$

4.7 Maximum Likelihood Learning for PPCA

Proposition 4.12 (PPCA ML Gradient). *The log-likelihood (cf. Proposition 3.4) is:*

$$\ell = -\frac{N}{2} \log |2\pi C| - \frac{N}{2} \text{Tr}[C^{-1}S] \quad (27)$$

where $C = \Lambda\Lambda^T + \psi I$.

The gradient with respect to Λ is:

$$\frac{\partial \ell}{\partial \Lambda} = \frac{N}{2} \left(-\frac{\partial}{\partial \Lambda} \log |C| - \frac{\partial}{\partial \Lambda} \text{Tr}[C^{-1}S] \right) = N(-C^{-1}\Lambda + C^{-1}SC^{-1}\Lambda) \quad (28)$$

Theorem 4.13 (PPCA ML Solution Structure). *At stationary points of the gradient (28), $SC^{-1}\Lambda = \Lambda$. This implies either:*

1. $\Lambda = 0$ (a minimum)
2. $C = S \Rightarrow \Lambda\Lambda^T = S - \psi I$. Since $\text{rank}(\Lambda\Lambda^T) \leq K$, we have $\text{rank}(S - \psi I) \leq K$, meaning S has $D - K$ eigenvalues equal to ψ , and Λ aligns with the space of remaining eigenvectors.
3. Taking the SVD $\Lambda = ULV^T$, the columns of U are eigenvectors of S (Proposition 4.3) with eigenvalues given by $l_i^2 + \psi$

The global ML solution is attained when Λ aligns with the K leading eigenvectors.

4.8 PPCA Latents

Proposition 4.14 (Posterior Distribution of Latents). *Given observation $\mathbf{x}_n$ from the PPCA model (Definition 3.1), the posterior distribution of the latent variables is:*

$$p(\mathbf{z}_n, \mathbf{x}_n) \propto \exp \left\{ -\frac{1}{2} [\mathbf{z}_n^T \Sigma^{-1} \mathbf{z}_n - 2\mathbf{z}_n^T \Sigma^{-1} \boldsymbol{\mu} + \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu}] \right\} \quad (29)$$

where:

$$\Sigma = (I + \Lambda^T \Psi^{-1} \Lambda)^{-1} = I - \beta \Lambda \quad (30)$$

$$\boldsymbol{\mu} = \Sigma \Lambda^T \Psi^{-1} \mathbf{x}_n = \beta \mathbf{x}_n \quad (31)$$

$$\beta = \Sigma \Lambda^T \Psi^{-1} \quad (32)$$

Corollary 4.15 (PPCA Projection). *The PPCA projection (using Proposition 4.14) is:*

$$\hat{\mathbf{x}}_n = \Lambda(I + \Lambda^T \Psi^{-1} \Lambda)^{-1} \Lambda^T \Psi^{-1} \mathbf{x}_n = \mathbf{x}_n - \Psi(\Lambda\Lambda^T + \Psi)^{-1} \mathbf{x}_n \quad (33)$$

This differs from the PCA projection (Definition 4.9). PPCA accounts for noise in the principal subspace. As $\psi \rightarrow 0$, the PPCA estimate approaches the PCA value.

5 Mixture Distributions

5.1 Definition

Definition 5.1 (Finite Mixture Distribution). A finite mixture distribution is a latent variable model (Definition 1.1) with a single discrete latent variable:

$$s_i \stackrel{\text{iid}}{\sim} \text{Discrete}[\boldsymbol{\pi}] \quad (34)$$

$$\mathbf{x}_i \mid s_i \sim P_{s_i}[\boldsymbol{\theta}_{s_i}] \quad (35)$$

Under the discrete distribution:

$$P(s_i = m) = \pi_m, \quad \pi_m \geq 0, \quad \sum_{m=1}^k \pi_m = 1 \quad (36)$$

Remark 5.2 (Mixture Motivation). Mixtures arise naturally when observations from different sources have been collated. They can also be used to approximate arbitrary distributions.

5.2 Mixture Likelihood

Proposition 5.3 (Mixture Probability). *The probability (density) at a single data point $\mathbf{x}_i$ under a finite mixture (Definition 5.1) is:*

$$P(\mathbf{x}_i) = \sum_{m=1}^k P(\mathbf{x}_i \mid s_i = m)P(s_i = m) = \sum_{m=1}^k \pi_m P_m(\mathbf{x}_i; \boldsymbol{\theta}_m) \quad (37)$$

The mixture distribution (density) is a convex combination (weighted average) of the component distributions (densities).

5.3 Approximation with Mixture of Gaussians

Remark 5.4 (Density Approximation). The component densities may be viewed as elements of a “basis” which can be combined to approximate arbitrary distributions. Given enough mixture components, we can model almost any density as accurately as desired, while only working with the well-known Gaussian form.

Example 5.5 (Non-Gaussian Approximations). Non-Gaussian densities (uniform, triangular, heavy-tailed) can be modeled as mixtures of Gaussians (Definition 5.1) with varying means, variances, and mixture weights.

5.4 Clustering with Mixture of Gaussians

Definition 5.6 (Clustering Interpretation). In clustering applications, the latent variable s_i of a mixture model (Definition 5.1) represents the (unknown) identity of the cluster to which the i -th observation belongs.

The latent distribution gives the prior probability of a data point coming from each cluster:

$$P(s_i = m \mid \boldsymbol{\pi}) = \pi_m \quad (38)$$

Data from the m -th cluster are distributed according to the m -th component:

$$P(\mathbf{x}_i \mid s_i = m) = P_m(\mathbf{x}_i) \quad (39)$$

Definition 5.7 (Responsibility). The posterior probability distribution for the cluster to which a data point belongs (by Bayes' rule applied to Definition 5.6) is:

$$P(s_i = m \mid \mathbf{x}_i) = \frac{P_m(\mathbf{x}_i)\pi_m}{\sum_{m'} P_{m'}(\mathbf{x}_i)\pi_{m'}} \quad (40)$$

This is called the responsibility of the m -th cluster for the i -th data point.

5.5 Mixture of Gaussians Likelihood

Definition 5.8 (MoG Model). Each component of a MoG is a Gaussian with mean $\boldsymbol{\mu}_m$ and covariance matrix Σ_m . The likelihood for n iid observations $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is:

$$p(\mathcal{D} \mid \{\boldsymbol{\mu}_m\}, \{\Sigma_m\}, \boldsymbol{\pi}) = \prod_{i=1}^n \sum_{m=1}^k \pi_m \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_m, \Sigma_m) \quad (41)$$

Proposition 5.9 (MoG Log-Likelihood). The log-likelihood of the MoG model (Definition 5.8) is:

$$\log p(\mathcal{D} \mid \{\boldsymbol{\mu}_m\}, \{\Sigma_m\}, \boldsymbol{\pi}) = \sum_{i=1}^n \log \sum_{m=1}^k \pi_m \frac{1}{\sqrt{|2\pi\Sigma_m|}} e^{-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_m)^T \Sigma_m^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_m)} \quad (42)$$

Note that the logarithm fails to simplify the component density terms. A mixture distribution does not lie in the exponential family, making direct optimization difficult.

5.6 Maximum Likelihood for Mixture Models

Proposition 5.10 (Mixture Likelihood Derivatives). The log-likelihood is:

$$\mathcal{L} = \sum_{i=1}^n \log \sum_{m=1}^k \pi_m P_m(\mathbf{x}_i; \theta_m) \quad (43)$$

The partial derivative with respect to θ_m is:

$$\frac{\partial \mathcal{L}}{\partial \theta_m} = \sum_{i=1}^n \frac{\pi_m}{\sum_{m'=1}^k \pi_{m'} P_{m'}(\mathbf{x}_i; \theta_{m'})} \frac{\partial P_m(\mathbf{x}_i; \theta_m)}{\partial \theta_m} = \sum_{i=1}^n r_{im} \frac{\partial \log P_m(\mathbf{x}_i; \theta_m)}{\partial \theta_m} \quad (44)$$

The partial derivative with respect to π_m is:

$$\frac{\partial \mathcal{L}}{\partial \pi_m} = \sum_{i=1}^n \frac{P_m(\mathbf{x}_i; \theta_m)}{\sum_{m'=1}^k \pi_{m'} P_{m'}(\mathbf{x}_i; \theta_{m'})} = \sum_{i=1}^n \frac{r_{im}}{\pi_m} \quad (45)$$

where r_{im} is the responsibility of cluster m for data point i (Definition 5.7).

5.7 MoG Derivatives

Proposition 5.11 (Gaussian Mixture Gradients). For a MoG (Definition 5.8) with $\theta_m = \{\boldsymbol{\mu}_m, \Sigma_m\}$, specializing the general derivatives (Proposition 5.10):

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\mu}_m} = \sum_{i=1}^n r_{im} \Sigma_m^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_m) \quad (46)$$

$$\frac{\partial \mathcal{L}}{\partial \Sigma_m^{-1}} = \frac{1}{2} \sum_{i=1}^n r_{im} [\Sigma_m - (\mathbf{x}_i - \boldsymbol{\mu}_m)(\mathbf{x}_i - \boldsymbol{\mu}_m)^T] \quad (47)$$

Remark 5.12 (Gradient-Based Learning). These equations can be used (with Lagrangian derivatives for π_m enforcing normalization (36)) for gradient-based learning, e.g., taking small steps in the gradient direction or using conjugate gradients.

6 K-Means Algorithm

Definition 6.1 (K-Means Algorithm). The K-means algorithm is a limiting case of the mixture of Gaussians (Definition 5.8). Take $\pi_m = 1/k$ and $\Sigma_m = \sigma^2 I$ with $\sigma^2 \rightarrow 0$. The responsibilities (Definition 5.7) become binary:

$$r_{im} \rightarrow \delta(m, \arg \min_l \|\mathbf{x}_i - \boldsymbol{\mu}_l\|^2) \quad (48)$$

with 1 for the component with the closest mean and 0 for all others.

Remark 6.2 (K-Means Iterations). The k-means algorithm iterates two steps:

1. **Assignment:** Assign each point to its closest mean:

$$r_{im} = \delta(m, \arg \min_l \|\mathbf{x}_i - \boldsymbol{\mu}_l\|^2) \quad (49)$$

2. **Update:** Update means to the average of their assigned points:

$$\boldsymbol{\mu}_m = \frac{\sum_i r_{im} \mathbf{x}_i}{\sum_i r_{im}} \quad (50)$$

This usually converges within a few iterations, though the fixed point depends on initial values for $\boldsymbol{\mu}_m$. The algorithm has no learning rate, but the assumptions are quite limiting.

7 EM Algorithm Preview

Remark 7.1 (EM for MoGs). Suppose we use the fractional responsibilities (Definition 5.7) from the full (non-limiting) MoG (Definition 5.8) but neglect their dependence on the parameters. We can solve for both $\boldsymbol{\mu}_m$ and Σ_m .

The EM algorithm for MoGs iterates:

1. **E-step:** Evaluate responsibilities (40) for each point given current parameters
2. **M-step:** Optimize parameters assuming responsibilities stay fixed:

$$\boldsymbol{\mu}_m = \frac{\sum_i r_{im} \mathbf{x}_i}{\sum_i r_{im}} \quad (51)$$

$$\Sigma_m = \frac{\sum_i r_{im} (\mathbf{x}_i - \boldsymbol{\mu}_m)(\mathbf{x}_i - \boldsymbol{\mu}_m)^T}{\sum_i r_{im}} \quad (52)$$

Although this appears ad hoc, it is a special case of a general algorithm guaranteed to increase the likelihood at each iteration.

7.1 Issues with Mixture Models

Several problems arise with these algorithms:

- Slow convergence for gradient-based methods
- Gradient-based methods may develop invalid covariance matrices
- Local minima; final configuration may depend on starting state
- Choosing k : Using likelihood alone is inadequate
- Singularities: Components with single data points have covariance approaching zero, causing likelihood to tend to infinity

8 Tools: Information Theory

8.1 Measuring Information

8.1.1 Information Content of a Random Variable

Consider a random variable X with distribution P .

- P expresses what we know before observing X
- How much information do we gain by observing X ?

Remark 8.1 (Information Content). By information content of X , we mean the difference in information between knowing P and knowing both P and $X = x$.

For the definition of information, think of:

- The distribution P as what we expect to happen
- The sample outcome $X = x$ as what actually happens

8.1.2 Heuristic Motivation

Suppose we sample $X = x$ from distribution P .

- If $P(x)$ is large: Small surprise; we have not gained much additional information
- If $P(x)$ is small: We have gained more information

Conclusions:

- Information in $X = x$ increases with $\frac{1}{P(x)}$
- Intuitively, information gain in two unrelated observations should be additive, so $\frac{1}{P(x)}$ itself is not useful

Definition 8.2 (Information). The information in observing $X = x$ under P is:

$$J_P(x) := \log \frac{1}{P(x)} = -\log P(x) \quad (53)$$

8.2 Shannon's Entropy

We consider discrete random variables (finite or countably infinite).

Definition 8.3 (Shannon Entropy). Let X be a discrete random variable with distribution P . The expected information (Definition 8.2) in a draw from P :

$$H[X] := \mathbb{E}_P[J_P(X)] = - \sum_{x \in \mathcal{X}} P(x) \log P(x) \quad (54)$$

is called the Shannon entropy of X , or the entropy for short. We use $H[X]$ and $H(P)$ interchangeably.

Proposition 8.4 (Basic Entropy Properties). *1. The entropy is non-negative: $H[X] \geq 0$*
2. $H(P) = 0$ means no uncertainty: $H(P) = 0 \Leftrightarrow P(x_0) = 1$ for some $x_0 \in \mathcal{X}$
3. If $\mathcal{X}$ is finite with d elements, the distribution with largest entropy is uniform U_d , with $H(U_d) = \log d$
4. $H(P)$ is concave as a function of P

8.3 Alternative Derivation

Suppose we define some measure $\mathcal{H}[X]$ of information in X . Instead of a definition, we postulate axioms that a meaningful measure should satisfy.

Axioms:

I. We should be able to “remove joint information”: $\mathcal{H}[X, Y] = \mathcal{H}[X] + \mathcal{H}[Y|X]$

This implies for independent variables: $X \perp Y \Rightarrow \mathcal{H}[X, Y] = \mathcal{H}[X] + \mathcal{H}[Y]$

II. $\mathcal{H}(P)$ should be continuous as a function of P

III. If we increase d , uncertainty increases: $\mathcal{H}(U_d) < \mathcal{H}(U_{d+1})$

Theorem 8.5 (Characterization of Entropy). *If a real-valued function $\mathcal{H}$ on $\mathcal{X}$ satisfies Axioms I–III, then:*

$$\mathcal{H}(P) = c \cdot H(P) \quad \text{for all } P \quad (55)$$

for some constant $c \in \mathbb{R}_+$, where $H(P)$ is the Shannon entropy (Definition 8.3).

Remark 8.6 (Axiomatic Uniqueness). Over the years, about a dozen different axioms for information measures have been proposed. Loosely speaking, any meaningful combination of two or three axioms leads to the same result (determines entropy up to scaling).

8.4 Example: Coding

Example 8.7 (Lossless Compression). Suppose we compress a text document losslessly.

Huffman Coding: A simple but efficient scheme:

1. Determine frequency of each word
2. Assign short code words to frequent words, long code words to rare words

This method is optimal for word-by-word substitution.

Information-theoretic problems: Suppose we know the distribution P of words in texts. Then:

1. What is the expected compression rate for a random document?
2. Does our encoder achieve the optimal expected rate for P ?

Theorem 8.8 (Source Coding Theorem (Shannon)). *Suppose we are given a distribution P on words or symbols and sample a string $X_1, \dots, X_n$ iid from P . Then for every $\varepsilon > 0$, there is a lossless encoder for which:*

$$H(P) \leq \mathbb{E} \left[\frac{1}{n} \cdot \text{length}(\text{encoding}(X_1, \dots, X_n)) \right] < H(P) + \varepsilon \quad (56)$$

for sufficiently large n , where $H(P)$ is the Shannon entropy (Definition 8.3).

Remark 8.9 (Entropy as Compression Limit). We can encode $X_1, \dots, X_n$ without information loss using $nH(P)$ bits on average. The entropy $H(P)$ is a lower bound: encoders achieving better rates have probability approaching 1 of information loss as $n \rightarrow \infty$.

Example 8.10 (English Text Compression). **Character-by-character:**

- Split text into characters, assign code to each
- Empirical character distribution has entropy ≈ 4.11 bit/character
- Not very effective: 27 characters, so $2^4 < 27 \leq 2^5$ means trivial 5-bit encoding

Word-by-word:

- Ranked word frequency in English follows Zipf distribution (parameter 1.5–2)
- Distribution highly concentrated on few common words (low entropy)
- Splitting into words achieves much better compression
- Common algorithms (e.g., Lempel-Ziv) split into substrings (not necessarily words)

8.5 Kullback-Leibler Divergence

8.5.1 Heuristic Motivation

Suppose we compare two distributions P and Q on $\mathcal{X}$.

- The entropy $H[Q] = \mathbb{E}_Q[J_Q(X)]$ (Definition 8.3) measures expected information gain (in terms of Q) from a random sample from Q
- Instead ask: How much information gain in terms of Q from a random sample from P ? Compute: $\mathbb{E}_P[J_Q(X)]$
- A measure of difference should vanish if $Q = P$. Since $P = Q$ means $\mathbb{E}_P[J_Q(X)] = H(P)$ (usually not 0), normalize by subtracting $H(P)$

Definition 8.11 (Kullback-Leibler Divergence). The Kullback-Leibler divergence or relative entropy of P and Q is:

$$\text{KL}[P\|Q] := \mathbb{E}_P[J_Q(X)] - H(P) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} \quad (57)$$

where J_Q is the information function (Definition 8.2) and $H(P)$ is the Shannon entropy (Definition 8.3).

Proposition 8.12 (KL Divergence Properties). 1. *Positive definiteness:* $\text{KL}[P\|Q] \geq 0$ and $\text{KL}[P\|Q] = 0 \Leftrightarrow P = Q$

2. $\text{KL}[P\|Q]$ is convex in both P and Q

8.5.2 Asymmetry

Remark 8.13 (KL Asymmetry). The KL divergence (Definition 8.11) is not symmetric: $\text{KL}[p\|q] \neq \text{KL}[q\|p]$ in general. It does not satisfy a triangle inequality. In particular, it is not a metric on probability distributions.

Asymmetry explanation:

- If p puts mass where q almost vanishes, then p/q is very large in that region
- Since both are normalized, there are other regions where p/q is small
- Because of the logarithm, these do not cancel out

Remark 8.14 (Pythagorean Property). Although KL divergence is not a metric, it exhibits geometric structure—in particular, it satisfies a property similar to Pythagoras' theorem.

8.6 Mutual Information

8.6.1 Heuristic Motivation

For a pair X, Y of random variables:

- How much information do they share?
- How much does observing X tell us about Y ?
- If X and Y contain no shared information, they are independent: $P(x, y) = P(x)P(y)$
- Idea: Compare actual joint distribution to independent case using KL divergence (Definition 8.11)

Definition 8.15 (Mutual Information). The mutual information of X and Y is:

$$I[X, Y] := \text{KL}[P(x, y)\|P(x)P(y)] = \sum_{x, y} P(x, y) \log \frac{P(x, y)}{P(x)P(y)} \quad (58)$$

Proposition 8.16 (Independence Characterization). *By the positive-definiteness of KL divergence (Proposition 8.12):*

$$I[X, Y] = 0 \quad \Leftrightarrow \quad X \perp Y \quad (59)$$

8.7 The Continuous Case

If the sample space $\mathcal{X}$ is uncountable (e.g., $\mathcal{X} = \mathbb{R}$), we consider densities p and q and substitute integrals for sums.

Definition 8.17 (Differential Entropy).

$$H[X] := - \int_{\mathcal{X}} p(x) \log p(x) dx \quad (60)$$

Note: Since p is a density, we can have $\log p(x) > 0$, so $H[X]$ can be negative. This is called differential entropy to distinguish it from (Shannon) entropy (Definition 8.3).

Definition 8.18 (KL Divergence (Continuous)). The continuous analogue of Definition 8.11:

$$\text{KL}[p\|q] := \int_{\mathcal{X}} p(x) \log \frac{p(x)}{q(x)} dx \quad (61)$$

Definition 8.19 (Mutual Information (Continuous)). The continuous analogue of Definition 8.15:

$$I[X, Y] := \int_{\mathcal{X}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx \quad (62)$$

Remark 8.20 (Continuous Properties). Unlike entropy, KL divergence (Definition 8.18) and mutual information (Definition 8.19) have the same properties as in the discrete case (Proposition 8.12 and Proposition 8.16).

8.8 Information Theory and Statistics

Points not discussed in detail:

- **Exponential families are maximum entropy models:** If T is a statistic with values in $\mathbb{R}^d$ and $\mathbf{t} \in \mathbb{R}^d$, the distribution maximizing entropy (Definition 8.3) under constraint $\mathbb{E}[T(X)] = \mathbf{t}$ is an exponential family with sufficient statistic T
- **Maximum likelihood minimizes KL divergence** (Definition 8.11) between empirical distribution and model
- **Variance, covariance, and χ^2 -statistic** can be regarded as first-order approximations to entropy, mutual information (Definition 8.15), and KL divergence
- Various methods can be derived by substituting information-theoretic for traditional statistical quantities
- **Example:** Independent component analysis (ICA) is a PCA-like dimension-reduction technique (cf. Definition 4.1) measuring independence via mutual information rather than covariance

9 Expectation Maximisation

(This section would be expanded in full lecture notes with detailed derivations, proofs, and examples of the EM algorithm.)